The Journal of Open Source Software

MQT Core: The Backbone of the Munich Quantum Toolkit (MQT)

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DOI: 10.xxxxxx/draft

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Submitted: 08 November 2024
Published: unpublished

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Summary

MQT Core is an open-source C++ and Python library for quantum computing that forms the backbone of the quantum software tools developed as part of the *Munich Quantum Toolkit (MQT, (Wille et al., 2024))* by the Chair for Design Automation at the Technical University of Munich. To this end, it consists of multiple components that are used throughout the MQT, including a fully fledged intermediate representation (IR) for quantum computations, a state-of-the-art decision diagram (DD) package for quantum computing, and a state-of-the-art ZX-diagram package for working with the ZX-calculus. Pre-built binaries are available via PyPI for all major operating systems and all modern Python versions. MQT Core is fully compatible with IBM's Qiskit 1.0 and above (Javadi-Abhari et al., 2024), as well as the OpenQASM format (Cross et al., 2022), enabling seamless integration with the broader quantum computing community.

Statement of Need

Quantum computing is rapidly transitioning from theoretical research to practice, with potential applications in fields such as finance, chemistry, machine learning, optimization, cryptography, and unstructured search. However, the development of scalable quantum applications requires automated, efficient, and accessible software tools that cater to the diverse needs of end users, engineers, and physicists across the entire quantum software stack.

The Munich Quantum Toolkit (MQT, (Wille et al., 2024)) addresses this need by leveraging decades of design automation expertise from the classical computing domain. Developed by the Chair for Design Automation at the Technical University of Munich, the MQT provides a comprehensive suite of tools designed to support various design tasks in quantum computing. These tasks include high-level application development, classical simulation, compilation, verification of quantum circuits, quantum error correction, and physical design.

MQT Core offers a flexible intermediate representation for quantum computations that forms the basis for working with quantum circuits throughout the MQT. The library provides interfaces to IBM's Qiskit (Javadi-Abhari et al., 2024) and the OpenQASM format (Cross et al., 2022) to make the developed tools accessible to the broader quantum computing community. Furthermore, MQT Core integrates state-of-the-art data structures for quantum computing, such as decision diagrams (Wille et al., 2023) and the ZX-calculus (Duncan et al., 2020; van de Wetering, 2020), that power the MQT's software packages for classical quantum circuit simulation (MQT DDSIM), compilation (MQT QMAP), verification (MQT QCEC), and more. As such, MQT Core has enabled more than 30 research papers over its first five years of development (Burgholzer et al., 2020; Burgholzer, Bauer, et al., 2021; Burgholzer, Raymond, et al., 2021; Burgholzer, Kueng, et al., 2021; Burgholzer, Ploier, et al., 2022a,

2022b; Burgholzer, Schneider, et al., 2022; Burgholzer & Wille, 2020a, 2020b, 2021; Grurl, Pichler, et al., 2023; Grurl et al., 2020, 2021; Grurl, Fuß, et al., 2023; Hillmich et al., 2021, 2020, 2022; Hillmich, Markov, et al., 2020; Hillmich, Kueng, et al., 2020; Peham et al., 2022b, 2022a, 2023b, 2023a; Peham, Brandl, et al., 2023; Sander et al., 2023; Schmid, Locher, et al., 2024; Schmid, Park, et al., 2024; Schneider et al., 2023; Wille et al., 2020, 2021, 2022, 2023; Wille & Burgholzer, 2022).

To ensure performance, MQT Core is primarily implemented in C++. Since the quantum computing community predominantly uses Python, MQT Core provides Python bindings that allow seamless integration with existing Python-based quantum computing tools. In addition, pre-built Python wheels are available for all major platforms and Python versions, making it easy to install and use MQT Core in various environments without the need for manual compilation.

Related Work

MQT Core builds on a rich history of research in quantum computing, design automation, and data structures. The design of its IR is heavily inspired by IBM's Qiskit (Javadi-Abhari et al., 2024), with which it has stayed compatible since `qiskit-terra` version 0.16.1 (released at the end of 2020). MQT Core remains one of the few libraries providing drop-in replacements for large parts of Qiskit's core data structures in C++. Alternative IRs, that come as part of larger quantum computing frameworks, include Quantinuum's C++-based t|ket⟩ (Sivarajah et al., 2021), LBNL's Python-based `bqskit` (Younis et al., 2021), Xanadu's MLIR-based `catalyst` (Ittah et al., 2024), and NVIDIA's MLIR-based `CUDA-Q` (*CUDA-q*, 2024).

The origin of the decision diagram package in MQT Core dates back to the seminal work on Quantum Multiple-Valued Decision Diagrams (QMDDs) (Niemann et al., 2016). It provides a state-of-the-art implementation of QMDDs that natively integrates with the MQT Core IR. Alternative types of quantum decision diagrams and related software packages include TDD's (Hong et al., 2022), Bit-Slicing Decision Diagrams (Tsai et al., 2021), and LIMDDs (Vinkhuijzen et al., 2023). In comparison to MQT Core, most of these libraries have remained academic prototypes and have not seen widespread adoption in the quantum computing community.

The ZX-diagram package in MQT Core is inspired by the `PyZX` library (Kissinger & Wetering, 2020) and the t|ket⟩ (Sivarajah et al., 2021) compiler. It provides an efficient C++ implementation of core ZX-calculus concepts and is tightly integrated with the MQT Core IR. Compared to other implementations, the ZX package in MQT Core is fine-tuned for verification use cases and provides dedicated support for handling qubit permutations as well as numerical inaccuracies that arise in practice.

Acknowledgements

The Munich Quantum Toolkit has been supported by the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation program (grant agreement No. 101001318), the Bavarian State Ministry for Science and Arts through the Distinguished Professorship Program, as well as the Munich Quantum Valley, which is supported by the Bavarian state government with funds from the Hightech Agenda Bayern Plus.

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